

1.5c

Sleep

Sleep loss, sleep disorders & dreams

What if we don't sleep enough?

After several nights of 5 hrs of sleep, we accumulate a sleep debt that can't be satisfied by one ^{low} sleep.

The brain keeps count of sleep debt for ≥ 2 weeks

Allowed to sleep unhindered, adults paying off sleep debt will sleep ≥ 9 hrs a night.

75% US students report < 8 hrs of sleep nightly

28% report falling asleep in class $\geq \frac{1 \text{ time}}{\text{week}}$

69% College students in one US study report "feeling tired" or "having little energy" on several days out of last 2 weeks

In two large studies, adolescents sleeping ≤ 5 hrs nightly, had 20% higher depression risk, 80% higher risk of suicidal thinking, compared to those with ≥ 8 hrs

REM processing of emotional experiences helps protect against depression.

later school start times consistently produced more sleep, better attendance, improved alertness, fewer car accidents

Lack of sleep can also increase weight gain.

- it increases ghrelin, a hunger-arousing hormone, and decreasing the opposite leptin.
- increase stress hormone cortisol → body make more fat
- enhancing limbic responses to food - increasing temptation

Sleep deprivation slows reactions & increases error on visual attention tasks

When especially drowsy, we may experience a 1-6 sec microsleep

What are the major sleep disorders?

Major sleep disorders

insomnia

narcolepsy

sleep apnea

sleep walking

REM sleep behavior disorder

Insomnia

$\frac{1}{5}$ adults

Ongoing difficulty falling or staying asleep

Causes chronic tiredness, increased risk of depression, obesity, hypertension, and arthritic & fibromyalgia pain

Narcolepsy

$\frac{1}{2000}$ adults

Sudden attacks of overwhelming sleepiness

risk of falling asleep at a dangerous moment; attacks usually last < 5 min, but can happen at the worst times

Sleep Apnea

$\frac{1}{20}$ adults; stoppage breathing repeatedly during sleep

causes fatigue and depression; associated with obesity

Sleep walking

$\frac{1-15}{100}$ of general population

repeated episodes of complex motor movement like walking in N3 sleep

REM sleep behavior disorder

age > 50 , $\frac{1}{50}$, $\frac{1}{100}$

acting out dreams

risks accidental injury

Insomniacs tend to overestimate the degree of severity of their insomnia.

Quick fixes like alcohol & sleep pills can aggravate the problem by reducing REM. Such aids can create tolerance.

Why do we dream?

REM dreams are vivid, emotional & bizarre.

~8/10 dreams are marked by ≥ 1 negative event or emotions.

$\frac{1}{10}$ in men, $\frac{1}{30}$ in women $\rightarrow$ 1/3 of dreams are sexual

Dream storylines incorporate traces of recent events

After trauma, people commonly report nightmares. For example, Palestinian children, after facing land-theft, displacement & genocide experience frequent trauma-related dreams.

Compared with nonmusicians, musicians report 2x as many dreams with music.

Blind people mostly dream nonvisually, but even people born blind sometimes 'see' in dreams

Anything that happens 5 min before sleep is typically lost from memory

Dreams may help sift, sort & fix the day's experience in our memory.

Brain scans confirmed link between REM sleep & memory.

Dreams may provide the sleeping brain with periodic stimulation, helps neural development

Some researchers propose dreams come from neural activation spreading up from brainstem.

Activation synthesis theory $\therefore$ dreams are brains attempt to synthesize random neural activity.

Personal Connection

When I decided to do something about my sleep issues (buy melatonin), my primary motivation was losing critical, limited, opportunities for brain development.

Some researchers see dreams as part of
brain maturation & cognitive development.

Deprived of REM, people reach REM quicker. When
allowed to sleep, they sleep like babies with increased
REM :: REM rebound.